

Lecture 10 Worksheet

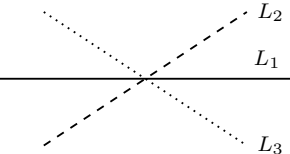
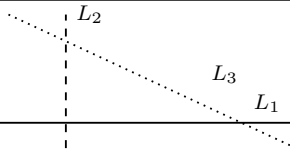
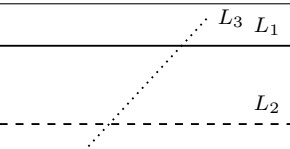
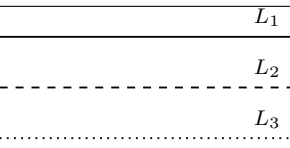
Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

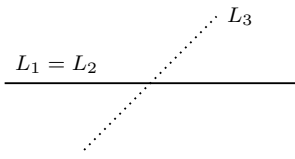
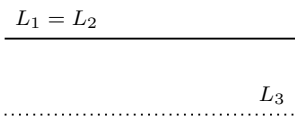
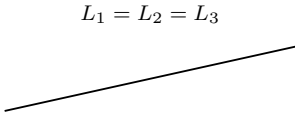
Activity 1: Three equations in two variables

Individual work.

For each picture below in $\mathbb{R}^2$, write three linear equations in x and y with the same arrangement. List the equations for L_1, L_2, L_3 in that order. The exact slopes and spacing need not match. Labels such as $L_1 = L_2$ mean that the lines coincide. Describe the points that satisfy all three equations. Each equation must describe a genuine line (i.e., either x or y must have a nonzero coefficient – so for example, $0 = 0$ or $0 = 5$ are not valid equations for this problem.)

Configuration	Your three equations	Common solution set
A 	L_1 : _____	_____
	L_2 : _____	_____
	L_3 : _____	_____
B 	L_1 : _____	_____
	L_2 : _____	_____
	L_3 : _____	_____
C 	L_1 : _____	_____
	L_2 : _____	_____
	L_3 : _____	_____
D 	L_1 : _____	_____
	L_2 : _____	_____
	L_3 : _____	_____

Activity 1: Three equations in two variables (continued)

	Configuration	Your three equations	Common solution set
E		$L_1 :$ _____ $L_2 :$ _____ $L_3 :$ _____	_____ _____ _____
F		$L_1 :$ _____ $L_2 :$ _____ $L_3 :$ _____	_____ _____ _____
G		$L_1 :$ _____ $L_2 :$ _____ $L_3 :$ _____	_____ _____ _____

Which of these pictures could represent a homogeneous system? Explain. No coordinate axes are shown, so you may choose where to place the origin.

Activity 2: Three planes in three-dimensional space

Group work.

Classify all possible configurations of three planes in $\mathbb{R}^3$. Planes may be parallel, may intersect, or may coincide. On the board, draw and label a picture of each configuration. For each picture:

- Label the planes P_1, P_2, P_3 , including any that coincide.
- Show where each pair of planes intersects, if they intersect.
- Describe the intersection of all three planes, and state whether the system has no solutions, one solution, or infinitely many.
- **Write three linear equations with numerical coefficients that produce the configuration.**

You can count configurations by thinking about whether planes could coincide, are parallel, and how they intersect. Changing angles, spacing, orientation, or plane names does not give a new case. Each equation must have at least one nonzero coefficient of x, y , or z .

Hint: there are eight configurations under these conventions.

Record your group's classification. Use the board for sketches and equations.

Case	Description of the configuration	Intersection of all three
1		
2		
3		
4		

Activity 2: Three planes in three-dimensional space (continued)

Record your group's classification (continued). Use the board for sketches and equations.

Case	Description of the configuration	Intersection of all three
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5

6

7

8

Which configurations could represent a homogeneous system? Explain. You may choose where to place the origin.

Discussion. If every pair of planes intersects, must all three share a point? How can you be sure your list includes every configuration?